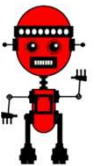


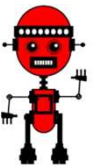
Linear Regression Machine Learning Algorithm



CONTENT



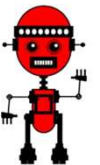
- Regression and types of regression
- Concept of Linear Regression
- Mathematics
- Terminology related to LR
- Some Prominent Applications of LR
- Evaluation Parameters
- Take away points



REGRESSION

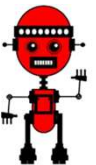
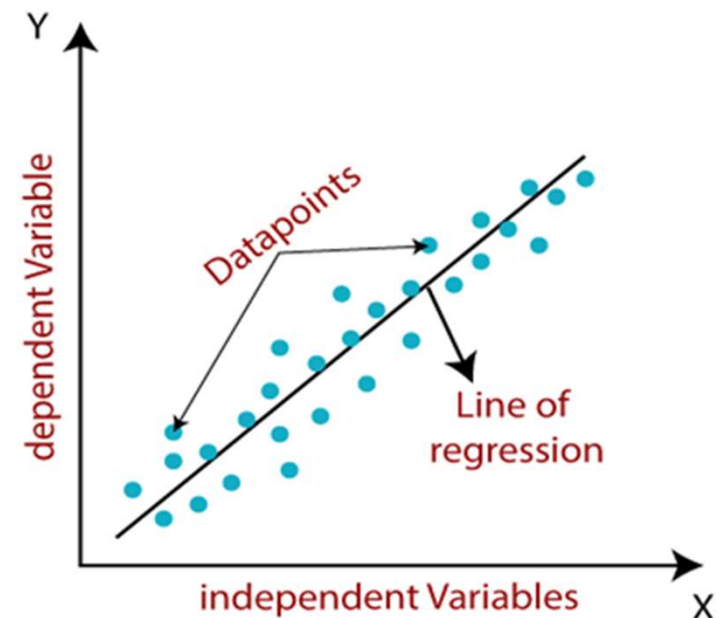
Used in finding the relationship between explanatory variables (or independent variables) and our variable of interest (or dependent variable).

Types of Regression



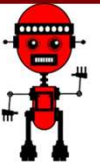
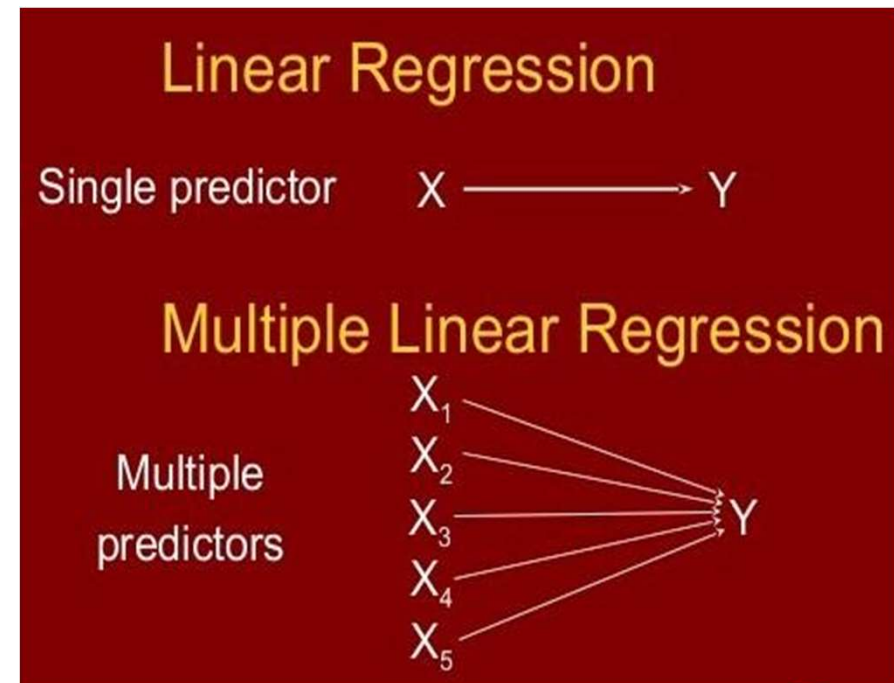
LINEAR REGRESSION

- It is supervised machine learning algorithm.
- Algorithm shows a linear relationship between a dependent (y) and one or more independent (x) variables, hence called as linear regression.
- Finds, how the value of the dependent variable is changing according to the value of the independent variable.



TYPES OF LINEAR REGRESSION

- **Simple linear regression**
 - Relationship between one feature and one outcome variable.
 - E.g. predicting income based upon age.
- **Multiple linear regression**
 - Relationship between more than one feature and one outcome variable.
 - E.g. predicting income based upon age and sex.



MATHEMATICS BEHIND LINEAR REGRESSION

For Simple Linear Regression:

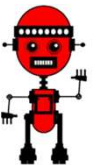
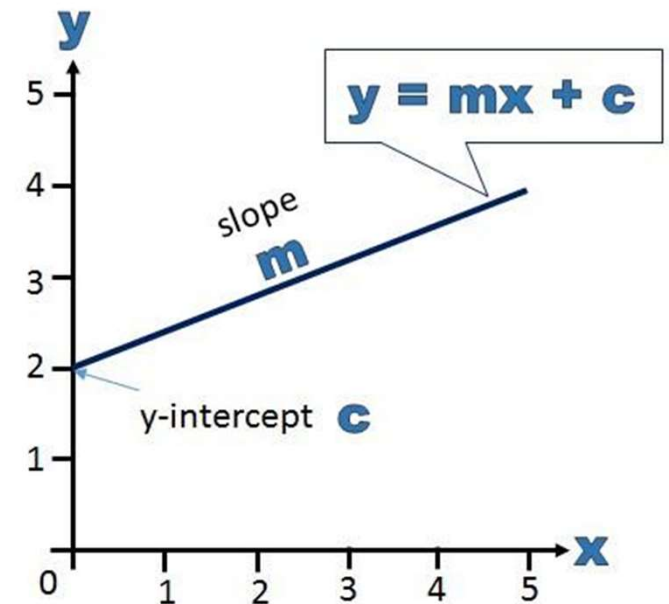
Here,

Y= Dependent Variable (Target Variable)

X= Independent Variable (predictor Variable)

c = intercept of the line (Gives an additional degree of freedom)

m = Linear regression coefficient (scale factor to each input value).



For Multiple Linear Regression:

$$Y = m_1x_1 + m_2x_2 + \dots + m_nx_n + c$$

Where,

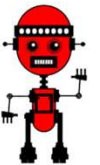
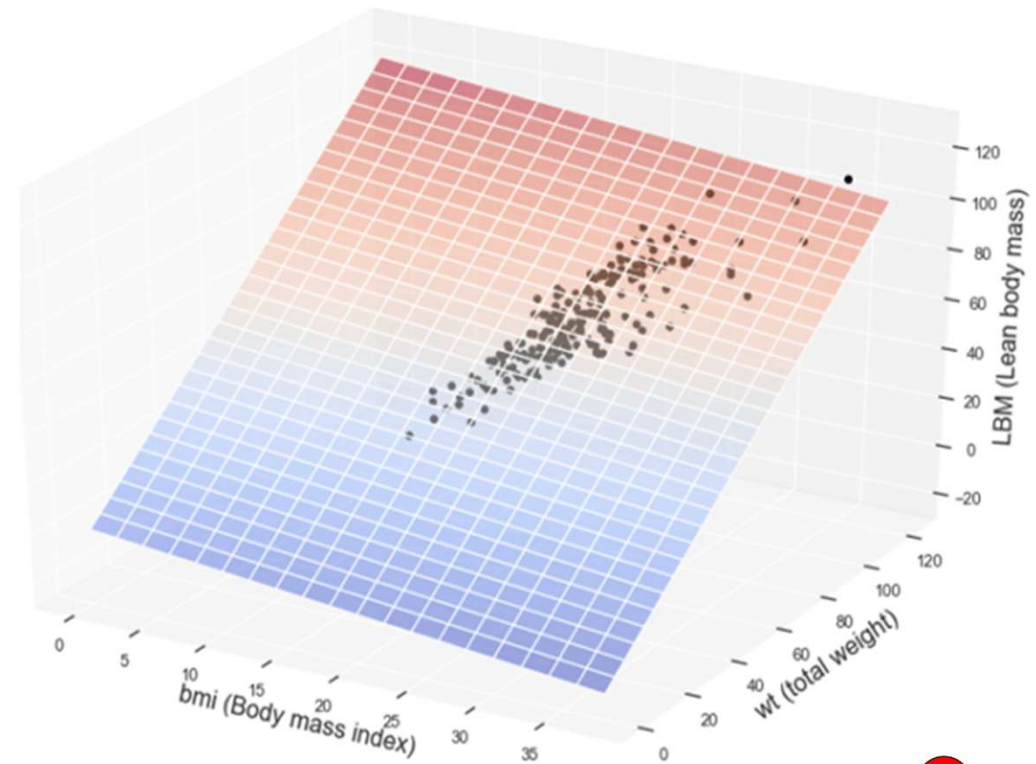
Y= Dependent Variable (Target Variable)

X= Independent Variable (predictor Variable)

n = Number of feature

m = Co-efficient

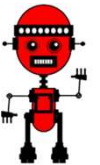
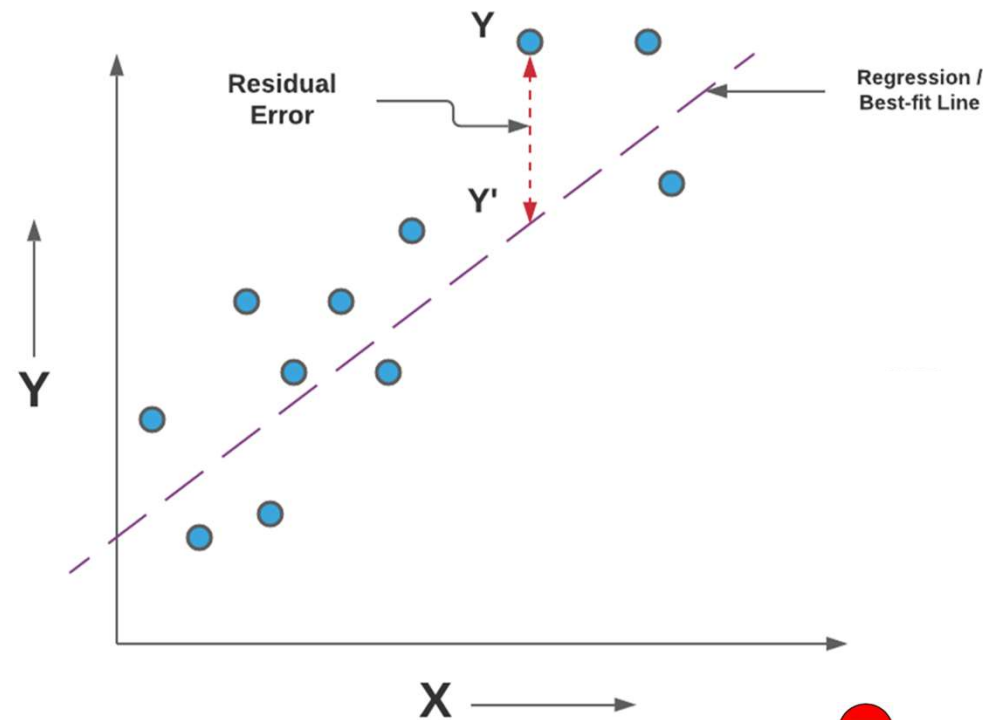
c = Y-intercept



Terminology related to LR

Cost function:

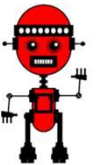
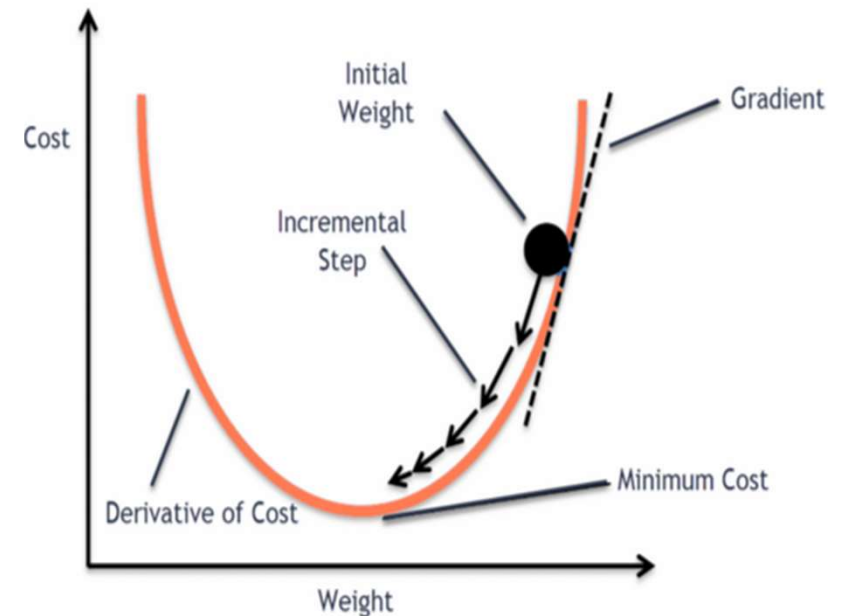
- Measure just how wrong the model is in finding a relation between the input and output
- How badly your model is behaving/predicting
- Obtained by subtracting the predicted values from actual values. The cost function will be the minimum of these error values.



Terminology related to LR

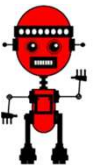
Gradient descent:

- Gradient descent is used to minimize the MSE by calculating the gradient of the cost function.
- Gradient Descent is an optimization algorithm for finding a local minimum of a differentiable function.
- Simply used to find the values of a function's parameters (coefficients) that minimize a cost function as far as possible
- Y - axis = Cost Function
- X- axis = Weight (m,b)



Some Prominent Applications of LR:

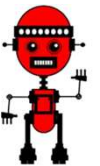
- Forecasting Sales
- Cash forecasting
- Analyzing survey data
- Stock predictions
- Predicting consumer behaviour



Evaluation Parameters

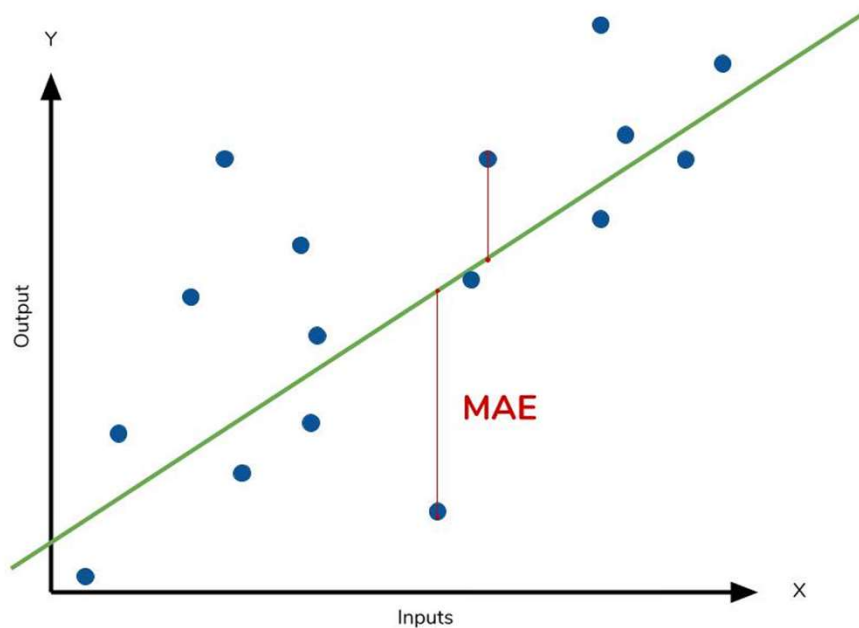
There are 4 main metrics for model evaluation in regression:

- Mean Absolute Error(MAE)
- Mean Square Error(MSE)
- Root Mean Square Error(RMSE)
- R Square/Adjusted R Square.



Mean Absolute Error(MAE):

- Shows the errors between actual values and values predicted by the model.



$$MAE = \frac{1}{n} \sum \left| y - \hat{y} \right|$$

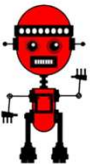
Divide by the total number of data points

Actual output value

Predicted output value

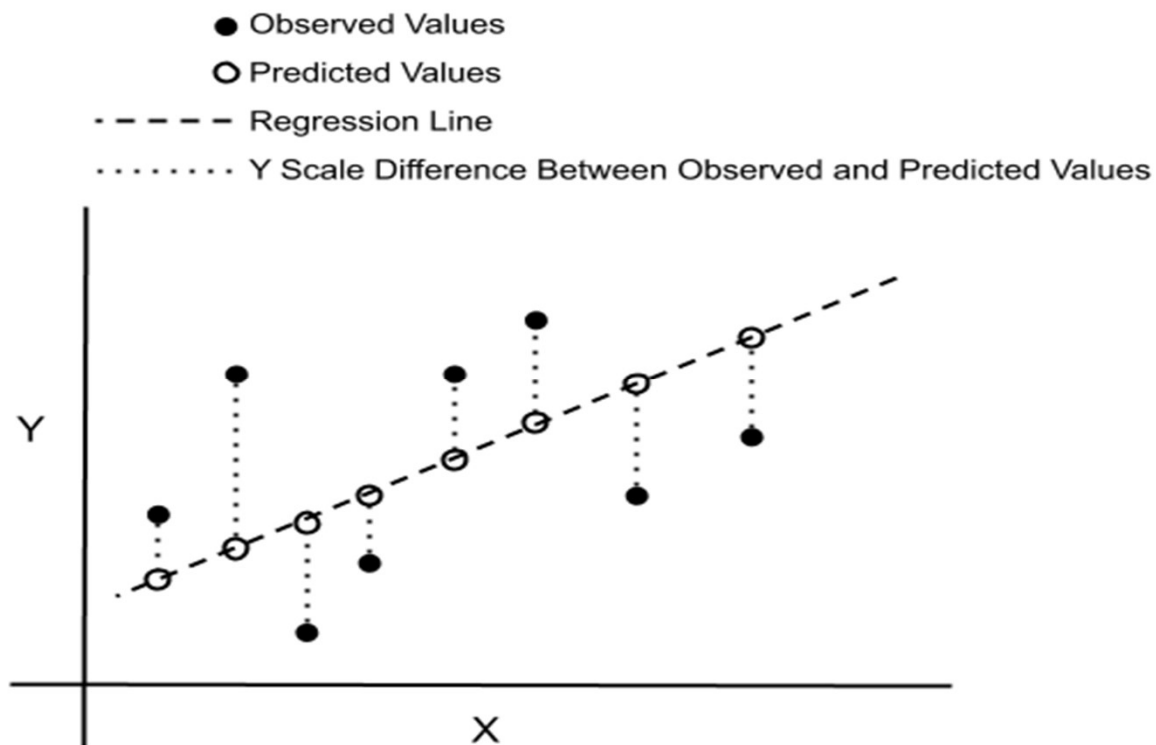
Sum of

The absolute value of the residual

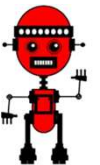


Mean Squared Error(MSE):

- Measures how close a regression line is to a set of data points.

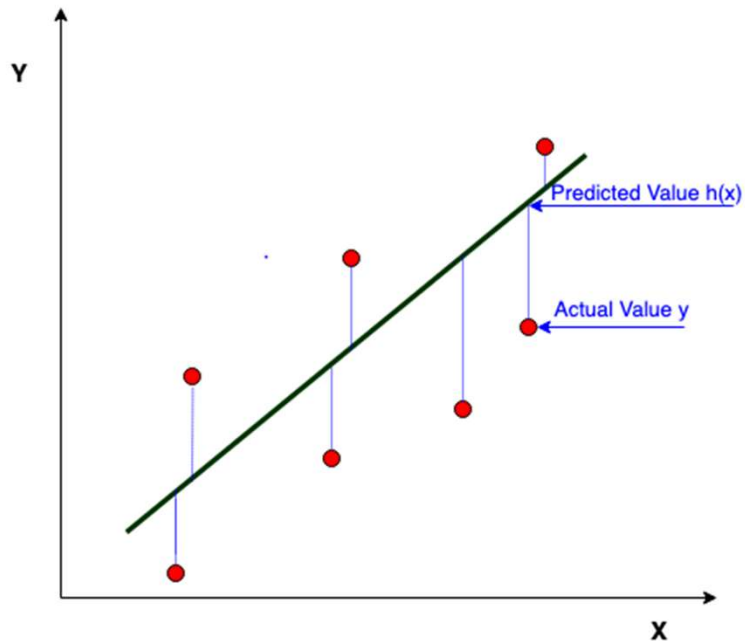


$$MSE = \frac{1}{n} \sum \left(\underbrace{y - \hat{y}}_{\text{The square of the difference between actual and predicted}} \right)^2$$

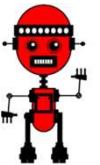


Root Mean Squared Error(RMSE):

- A simple square root of mean squared error.

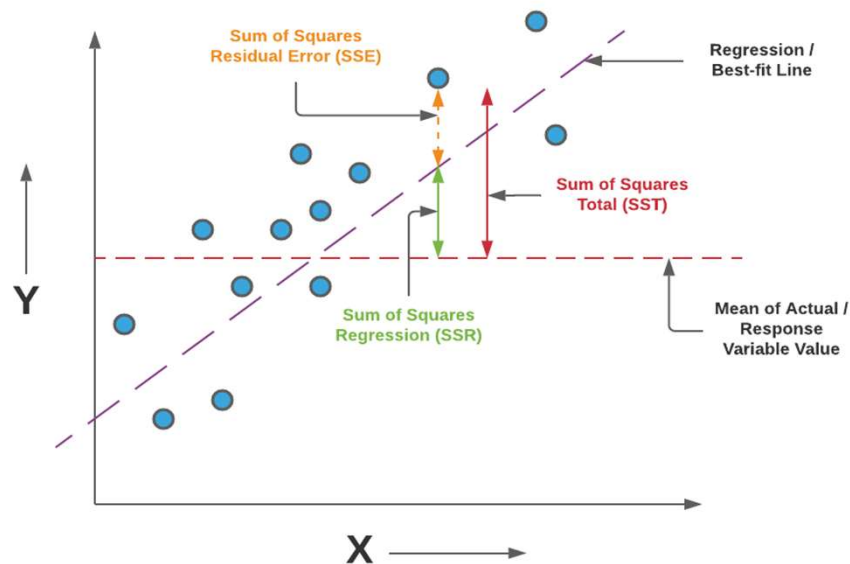


$$RMSE = \sqrt{\frac{\sum_{i=1}^n (y_i - \hat{y}_i)^2}{n}}$$

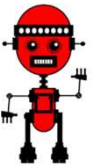


R Squared:

- Statistical measure in a regression model that determines the proportion of variance in the dependent variable that can be explained by the independent variable.

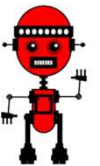


$$R^2 = 1 - \frac{\text{Sum Squared Regression Error} \rightarrow SS_{Regression}}{\text{Sum Squared Total Error} \rightarrow SS_{Total}}$$



Summary of data set

- Medical Premium prediction data set has 11 features and 986 rows.
- Strength of correlation between PremiumPrice(target variable) and Age is .7(positive) which is the highest correlation we got from the correlation matrix.It indicates that both are directly proportional.



Take Away Points

- Data has 11 null values in Age feature which is replaced with mean of Age.
- Blood Pressure Problems has 12 null values which is replaced with mode of Blood Pressure Problems because feature is numerical categorical in nature.
- After filling missing values ,correlation between Age and Premium Price remains same that means our imputation does not disturbed the distribution of data.
- No Duplicate values in dataset.
- Used linear regression because dependent variable is continuous in nature.
- Model got **70%** r^2_score , **12821966** MSE, **2624** MAE, **3580** RMSE.

